

Verification Model for Agentic Aircraft Design Workflows

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Project Context

Engineering design tools such as OpenMDAO, OpenConcept, OpenAeroStruct, pyCycle, and related aircraft-design codes are powerful but difficult to use correctly. They require the engineer to understand the assumptions, inputs, outputs, units, solver behavior, and limitations of each tool. The long-term research goal is to understand how **agentic AI systems** can help engineers run these tools, interpret their outputs, and detect when results are unreasonable.

For this summer project:

Develop a verification model that uses retrieval-augmented generation, historical aircraft data, and textbook equations to check whether aircraft design results are reasonable.

The computational test case will be the [OpenConcept King Air C90GT example](#). This example defines a custom King Air model using OpenConcept components for aerodynamics, weight, propulsion, and mission analysis. The script sets mission parameters such as climb rate, cruise speed, cruise altitude, mission range, and payload, then runs an OpenMDAO/OpenConcept analysis.

Project Objectives

The main objective is to build a **verification model** for aircraft design outputs.

The verification model should answer questions like:

- Is this predicted aircraft weight reasonable?
- Is the mission fuel burn reasonable?
- Is the cruise lift-to-drag ratio reasonable?
- Is the range reasonable for this aircraft class?
- Are the OpenConcept outputs consistent with historical aircraft data?
- Are the outputs consistent with textbook preliminary design equations?

- Can the agent explain what it checked and why?

By the end of the summer, the student should demonstrate a pipeline that:

1. Runs the OpenConcept King Air C90GT example.
2. Extracts key input/output quantities.
3. Uses a retrieval system to find relevant historical aircraft data.
4. Uses preliminary design equations to compute independent estimates.
5. Compares OpenConcept outputs against retrieved data and equations.
6. Flags unreasonable outputs.
7. Produces a traceable verification report for a human engineer.

Aircraft Information

Computational Example

OpenConcept King Air C90GT example

This is the baseline aircraft design case for the project. The example imports OpenMDAO, OpenConcept aerodynamics, weights, propulsion, mission analysis, and King Air aircraft data. It includes a `run_kingair_analysis()` function that sets up the problem, runs the model, and optionally produces plots.

OpenConcept methodology [paper](#)

Use this as background for understanding what OpenConcept is intended to do and how it supports conceptual aircraft design and mission analysis.

Historical Aircraft Data Sources

Sadraey, Aircraft Design: A Systems Engineering Approach

Focus on tables in Chapters 4, 5, 6, and 10. These contain scattered but useful aircraft-design data and empirical guidance. You might need to create a model similar to what is described in [this paper](#).

FAST AEROBASE

[AEROBASE](#) is a public aircraft and engine database from Gökçin Çınar's group. It contains data for over 400 aircraft and 200 engines, including aircraft weights, engine data, thrust-to-weight or power-related metrics, TSFC, and lift-to-drag ratio trends.

Empirical Correlation Models

[This source](#) provides data-driven regression and correlation models for aircraft design parameters. These models should be treated as an independent verification pathway.

Equation Sources

Nita dissertation: Preliminary Design and Optimization

Use Chapter 2 as a source for preliminary aircraft design equations.

Light Civil Turboprop Airplane Take-Off Weight Preliminary Design Estimation Method

This source is especially relevant because it focuses on light civil turboprop aircraft and includes King Air C90 GTx as one of the tested aircraft.

Week	Goal	Tasks	Deliverables
1	Create a clean baseline run of the King Air C90GT example.	- Clone OpenConcept - Run the King Air C90GT example. - Identify major OpenConcept/OpenMDAO components in the script. - Run KingAirC90GT.py directly. - Save all plots and numerical outputs. - Extract the key input and output variables. - Create an input/output schema.	- log showing the King Air example ran. Explain - what the model computes - what mission is being flown - what the main inputs/outputs mean - what quantities need verification
2	Run the same OpenConcept case through the agentic/MCP workflow.	- Use Alex's MCP tools to run the same King Air example. - Instructions are available here (Claude Code, Codex, Claude.ai): https://mcp.lakesideai.dev/#connect - Alternatively, the OpenConcept server can be run locally by cloning https://github.com/muroc-aero/the-hangar and opening a coding agent in the directory - Use `/mcp` with the agent, or settings in claude.ai to check the configuration - Compare direct execution vs agent execution. - Identify whether the agent changes anything. - Record all agent actions.	- direct-run vs agent-run comparison table - agent execution log
3	Define exactly what needs to be verified.	- Map OpenConcept variables to aircraft-design concepts. - Define units and expected physical ranges. - Identify which outputs can be compared against historical data. - Identify which outputs can be checked using equations.	- verification schema table - list of required checks - list of quantities not currently checkable
4	Gather the data & equations for the verification model.	- Identify all data sources. - Extract relevant data fields from Sadraey, AEROBASE, and other aircraft data. - Create a metadata schema for aircraft data. - Examples: - From FAST-GA [yaml, xml] - From RCAIDE [py] - Classify the King Air C90GT aircraft type.	- aircraft data schema - initial aircraft data table - source inventory - list of comparable aircraft
5	Build a minimum viable retrieval system.	- Implement a retrieval workflow. - Given an OpenConcept result, retrieve relevant comparison data. - Test retrieval on King Air-related queries. - Evaluate whether retrieved results are relevant.	- retrieval test set - retrieval evaluation table Metrics - retrieval relevance - parameter coverage - source coverage - unit consistency - aircraft-class consistency

			Example query: “Retrieve aircraft similar to the King Air C90GT and compare cruise speed, range, MTOW, OEW, payload, and engine power.”
6	Build independent equation-based checks.	• Implement equations. ◦ Start with simple, explainable equations. • Test equations against known aircraft data. • Compare equation outputs to retrieved aircraft data.	• equation documentation • test cases using real aircraft
7	Combine retrieval and equation tools into a verification model.	• For each key OpenConcept output, run retrieval comparison. • For each key OpenConcept output, run equation comparison where possible. • Define pass/fail thresholds. • Define confidence levels.	• OpenConcept verification report for King Air C90GT • list of checks that passed, failed, or require human review
8	Test whether the verification model can catch bad or unrealistic designs and test it on a different aircraft.	• Deliberately perturb the OpenConcept case. • Create unrealistic cases: ◦ excessive payload ◦ unrealistic range ◦ unreasonable cruise altitude ◦ unrealistic cruise speed ◦ incorrect fuel burn ◦ inconsistent aircraft weight • Run verification model on each case. • Have OpenConcept design a different aircraft: Caravan.	• bad-case dataset • detection results • failure analysis
9	Make the system understandable to a human engineer.	• Create a structured audit trail. • Log each retrieved source. • Log each equation. • Log each comparison. • Log each agent decision. • Create a human-readable verification report.	• traceability log • verification report template • example report for King Air C90GT • example report for one unrealistic case
10	Create and verify a new conceptual aircraft design Eco-Caravan: a more efficient, future-looking aircraft inspired by the Cessna Caravan / King Air-class turboprop design space.	• Have OpenConcept now make a hybrid-electric version based on powertrain architecture in this paper	• Is the design physically reasonable? • Which historical aircraft is it closest to? • Which assumptions are most uncertain? • Which outputs need higher-fidelity analysis? • What does the verification model flag?

Example Schema

OpenConcept Variable	Description	Unit	Verification method
fuel_burn	Mission fuel burn	lbs	Retrieval + Breguet range equation (Eq. x.x from xyz)
	Operating empty weight	lbs	Retrieval + weight regression equation (Eq. x.x from xyz)
	L/D		

Example Verification Table

OpenConceptVariable	OpenConcept Value	Retrieved value	Equation estimate	Pass/fail	Explanation
fuel_burn					

Agentic AI Resources

RAG

- **RAG Checkers** such as those from [Amazon](#)
- Representing information as knowledge graphs and [reasoning using LLMs](#)